MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

ACTUATORS, ELECTROMECHANICAL, LINEAR

TO ALL HOLDERS 0F MIL-STD-35-73(MI)

1. The following pages of MIL-STD-35-73(MI) have been revised and supersede the pages listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
v	15 January 1975	v	10 May 1974
vi	15 January 1975	vi	10 May 1974
vi i	15 January 1975	vi i	10 May 1974
viii	15 January 1975	viii through x	10 May 1974
5	15 January 1975	5	10 May 1974
6	15 January 1975	6	10 May 1974
7	15 January 1975	7	10 May 1974
8	15 January 1975	8 through 18	10 May 1974
19	15 January 1975	19 and 20	10 May 1974
21	15 January 1975	21 through 23	10 May 1974
24	10 May 1974	Reprinted without change	
35	15 January 1975	35 through 40	10 May 1974
41	15 January 1975	41 and 42	10 May 1974
43	15 January 1975	43	10 May 1974
44	10 May 1974	Reprinted without change	
101	15 January 1975	101 through 132	10 May 1974
145	15 January 1975	145 through 149	10 May 1974

2. Retain this notice page and insert before the Table of Contents.

3. Holders of MIL-STD-35-73(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or cancelled.

Custodians:
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Review Activities:
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SECTION 1.2

SPECIFICATION REQUIREMENTS INSTRUCTIONS

1.2.1 General

The purpose of these instructions is to establish the procedures to be followed, and the format to be used, for submission of requirement to obtain a computer generated procumbent document.

The requester will find it impossible to complete an SRS using only this dash numbered part of MIL-STD-35. Information which is standard for all generic families has been included in the master document at the issuance of MIL-STD-35C. This information is not repeated for each dash numbered part. The following list contains the subjects which are pertinent to every family unless otherwise excepted.

General Description of AEDPS
Part Numbering
Specification Requirements Sheet
Writing Methodology
Mandatory Specification Requirements
General Specification Requirements
Dimensions and Drawings
Applicable Documents
Materials
Finishes
Inspections
Single and Multi-Item Specification (Tabulation)
Sources
Search
General Instructions for SRL
Tables A through S

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1.2.2 Approved Item Name Code

The item name code for an actuator, to be entered on the SRS with Code 1007, is selected from the Item Name Code column of the following list.

Approved Item Name	Item Name Code
Actuator, Electromechanical, Linear	10957
Unapproved Names	77777

1.2.3 Figure Number

Code 1009 must show the figure number selected from Section 1.3.

1.2.4 Dimensions

MIL-STD-35 must be used in conjunction with this dash numbered part to completely delineate the dimensions for an actuator.

The figures and corresponding legends, representing different configurations of actuators, are located in the rear of this document. The figure number (representing the selected figure) is mandatory and shall be entered on the SRS with Code 1009. The figure legends in this document (AA, AB, AC, etc.) representing dimensions shall be employed with corresponding legends in Table A of MIL-STD-35.

The remaining instructions for dimensioning actuators are in MIL-STD-35.

1.2.5 Tabulation

The explanation of, and the instructions for using tabulation are found in MIL-STD-35. Candidates for tabulation for this part are the dimensional codes shown in Table A of MIL-STD-35 and the following.

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TABULATION CANDIDATES FOR LINEAR ACTUATORS

Characteristic Code	Title
4400	Minimum Starting Current (amperes)
4402	Maximum Starting Current (amperes)
4404	Minimum Starting Current (milliamperes)
4406	Maximum Starting Current (milliamperes)
4408	Minimum Operating Current (amperes)
4410	Maximum Operating Current (amperes)
4412	Minimum Operating Current (milliamperes)
4414	Maximum Operating Current (milliamperes)
4416	Minimum Operating Stroke (inches)
4418	Maximum Operating Stroke (inches)
4420	Minimum Operating Rate (inches per second)
4422	Maximum Operating Rate (inches per second)
1562	Rated Load (pounds)
1564	Rated Load (amperes)
1566	Rated Load Both Aiding and Opposing (pounds)
1568	Rated Load Both Aiding and opposing (amperes)
1570	Maximum Operating Load (pounds)
1572	Maximum Operating Load (amperes)
1574	Stall Load (pounds)
1576	Stall Load (amperes)
1620	Operating Stroke Adjustments (inches)
1644	Maximum Actuator Weight (ounces)
1646	Maximum Actuator Weight (pounds)
1750	Rated Life (number of full stroke cycles)

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NOTE: Paragraphs 1.2.6, 1.2.7, and 1.2.8 have been deleted.

1.2.9 Applicable Documents

In addition to the documents contained in the Applicable Document List of MIL-STD-35, the documents that follow shall be entered on the SRS with appropriate codes from Table B of MIL-STD-35.

Documents	Codes or Other Conditions
MIL-STD-130	Mandatory
MIL-STD-461	2512 and 2514 (Conducted Interference) and 2530 (Radiated Interference)
MIL-STD-462	2504 and 2506 (Conducted Interference) and 2524 (Radiated Interference)
MIL-STD-704	1658 (Input Power)
MIL-STD-810	When a characteristic code specifying a test method in MIL-STD-810 is entered on the SRS.
MIL-M-7969	1674 and 1676 (Motors)
MIL-M-8609	1670 and 1672 (Motors)
MS16624	1740 (Retaining Rings)
MS16625	1740 (Retaining Rings)
MS20995	1726 (Locking of Parts)
MS33540	1726 (Locking of Parts)
MS33547	1748 (Spring Pins)
NAS561	1774 (Spring Pins)

NOTE: Paragraphs 1.2.10 through 1.2.19 have been deleted.

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1.2.20 Design Characteristics

It is mandatory to specify design characteristics on the SRS with at least one of the following.

1.2.20.1 Starting Current

Starting current may be specified in amperes or in milliamperes with one of the following:

a. Amperes. Write in the minimum value (amperes) with Code 4400, or write in the maximum value (amperes) with Code 4402, or write in both the minimum and the maximum value (amperes) with Codes 4400 and 4402, respectively.

b. Milliamperes. Write in the minimum value (milliamperes) with Code 4404, or write in the maximum value (milliamperes) with 4406, or write in both the minimum and the maximum values (milliamperes) with Codes 4404 and 4406, respectively.

1.2.20.2 Operating Current

Operating current may be specified in amperes or in milliamperes with one of the following:

a. Amperes. Write in the minimum value (amperes) with Code 4408, or write in the maximum value (amperes) with Code 4410, or write in both the minimum and maximum values (amperes) with Codes 4408 and 4410, respectively.

b. Milliamperes. Write in the minimum value (milliamperes) with Code 4412 or write in the maximum value (milliamperes) with Code 4414, or write in both the minimum and maximum values (milliamperes) with Codes 4412 and 4414, respectively.

1.2.20.3 Operating Stroke

Operating stroke may be specified by writing in the minimum value (inches) with Code 4416, or by writing in the maximum value (inches) with Code 4418, or by writing in both the minimum and the maximum values (inches) with Codes 4416 and 4418, respectively.

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1.2.20.4 Operating Rate

Operating rate may be specified by writing in the minimum value (inches per second) with Code 4420, or by writing in the maximum value (inches per second) with Code 4422, or by writing in both the minimum and the maximum values (inches per second) with Codes 4420 and 4422, respectively.

1.2.20.5 Maximum Overtravel

Maximum overtravel may be specified by writing in the maximum value (inches) with Code 4424.

1.2.20.6 Maximum End Play

Maximum end play may be specified by writing in the maximum value (inches) with Code 4426.

NOTE: Paragraphs 1.2.21 and 1.2.22 have been deleted.

1.2.23 Rated Loads

Rated Load or Rated Load (both aiding and opposing) the stroke cycle may be specified on the SRS by using one of the following:

a. Rated Load. Write in the load value (pounds) with Code 1562. The load current may also be specified, if desired, by writing in the value (amperes) with Code 1564.

b. Rated Load (both aiding and opposing). Write in the load value (pounds) with Code 1566. The load current may also be specified, if desired, by writing in the current value (amperes) with Code 1568.

1.2.24 Maximum Operating Load

Maximum operating load may be specified on the SRS by writing in the maximum load value (pounds) with Code 1570. The load current may also be specified, if desired, by writing in the value (amperes) with Code 1572.

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1.2.25 Stall Load

Stall load may be specified on the SRS by writing in the load value (pounds) with Code 1574. The load current may also be specified, if desired, by writing in the value (amperes) with Code 1576.

1.2.26 Static Limit Load

Static limit load may be specified on the SRS either as a multiple of rated load or as a percent of the maximum operating load by using one of the following:

a. Multiple of Rated Load. Write in the multiplier value (number) with Code 1580 to specify that the actuator shall withstand, without any permanent deformation, a static load of (Code 1580) times the rated load.

b. Percent of Operating Load. Write in the value (percent) with Code 1582 to specify that the actuator shall withstand, without any permanent deformations, a static load equivalent to (Code 1582) percent of the maximum operating load.

In addition to a. or b. above, the following may also be used: Enter an "X" with Code 1584 to specify that the actuator is not required to operate under static limit load.

1.2.27 Ultimate Static Load

Ultimate static load may be specified on the SRS either as a percent of static limit load or in pounds by using one of the following:

a. Percent of Static Limit Load. Write in the value (percent) with Code 1590 to specify that the actuator shall withstand, without structural failure, an ultimate static load of (Code 1590) percent of the static limit load.

b. Pounds. Write in the value (pounds) with Code 1592 to specify that the actuator shall withstand, without structural failure, an ultimate static load of (Code 1592) pounds.

In addition to a. or b. above, one or more of the following may also be used:

(1) Write in the value (inches) with Code 1594 to specify that structural failure shall not occur with the actuator extended (Code 1594) inches.

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a. If the first test method is desired, use the first test method code with an "X" to specify testing per the specification referenced with Code 1006.

b. If the second test method is desired, use the second test method code and complete the sentence with the specified exceptions, for testing per the referenced specification with exceptions. The exceptions stated-must be limited to 189 characters, including punctuation and spaces.

c. If testing is to be detailed, one or more of the test method codes should be used to specify test conditions. All codes marked with an asterisk "*" must be used when employing this third alternate.

If the specification referenced with Code 1006 specifies a MIL-STD-81O test method and exceptions are to be taken, the second test method code (see b. above) should be used if the exceptions can be stated within the 189 character limitation; the third test method (see c. above) should be used if the exceptions cannot be stated within the limitation.

1.2.58.2 Requirements

If the requirements specified in the specification shown with Code 1006 are to be used, the first requirements code should be shown, with an "X" on the SRS. If exceptions to the requirements specified in the specification shown with Code 1006 are to be made, the second requirements code should be shown with exceptions written on the SRS to complete the sentence. If the requirements are to be changes in parameter values as a result of the environmental test, one or more of the remaining requirements codes with the applicable fill-in values must be shown. When the third requirements code is shown, the requirements that there be no evidence of physical damage or degradation of performance will be included automatically.

Custodians:
Army-MI

Review Activities:
Army-AT
Army-AV
Army-EL
Army-ME
Army-MR
Army-SG
Army-WC

User Activity:
Army-MU

Preparing Activity:
Army-MI
Project EDS-A099

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SPECIFICATION REQUIREMENTS LIST (SRL)

REF PARA	CHARACTERISTIC NAME	CODE
1.2.1	Enter on SRS, requirements from MIL-STD-35.	
	NOTE: Reference paragraphs 1.2.12 through 1.2.19 have been deleted. These characteristic names and codes are now contained in MIL-STD-35.	
1.2.20	*Design Characteristics (Select one or more and enter value)	
1.2.20.1	Starting current	
	Minimum (amps)	4400
	Maximum (amps)	4402
	or	
	Minimum (milliamps)	4404
	Maximum (milliamps)	4406
1.2.20.2	Operating current	
	Minimum (amps)	4408
	Maximum (amps)	4410
	or	
	Minimum (milliamps)	4412
	Maximum (milliamps)	4414
1.2.20.3	Operating stroke	
	Minimum (inches)	4416
	Maximum (inches)	4418

INDICATES MANDATORY REQUIREMENTS
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SPECIFICATION REQUIREMENTS LIST (SRL)

REF PARA	CHARACTERISTIC NAME	CHARACTERISTIC CODE
1.2.20.4	Operating rate	
	Minimum (inches per second)	4420
	Maximum (inches per second)	4422
1.2.20.5	Maximum overtravel (inches)	4424
1.2.20.6	Maximum end play (inches)	4426
	Note: If the Requirements and Quality Assurance Provisions in the specification shown with Code 1006 meet the user's requirements, Characteristic Codes without an asterisk "*" in the far left edge should not be used. The generated document will then specify all of the requirements and tests of the referenced specification.	
1.2.23	Rated Loads	
	Pounds.	1562
	Amperes	1564
	or	
	Rated Loads (both aiding and opposing)	
	Pounds	1566
	Amperes	1568
1.2.24	Maximum Operating Load	
	*Pounds	1570
	Amperes	1572

*INDICATES MANDATORY REQUIREMENTS
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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

REF. PARA.	CHARACTERISTIC NAME	CODE
1.2.25	Stall Load	
	*Pounds	1574
	Amperes	1576
1.2.26	Static Limit Load (Select one)	
	Static limit load (times the rated load)	1580
	or	
	Static limit load (percent of maximum operating load)	1582
	Operating under static limit load not required	1584
1.2.27	Ultimate Static Load without Structural Failure (Select one)	
	Ultimate static load (percent of static limit load)	1590
	or	
	Ultimate static load (pounds)	1592
	Actuator extension (inches)	1594
	Component deformation permitted	1596
	Inoperative failure permitted	1598

*INDICATES MANDATORY REQUIREMENTS

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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

REF. PARA.	CHARACTERISTIC NAME	CODE
1.2.58.1 Cont	Measurements after test (Select one or more)	
	Starting current	2470
	Operating current	2472
	Operating stroke	2474
	Operating rate	2476
	Overtravel	2478
	End play	2480
1.2.58.2	Requirement (Select one) (Mandatory if either Code 2450, 2452, or 2454 is selected)	
	Requirements per Specification shown with Code 1006	2482
	or	
	Requirements per Specification shown with Code 1006 except ____.	2484
	or	
	Requirements as follows:	
	*No evidence of physical damage or degradation of performance	2485
	Pretest requirements per Design Characteristics	2486
	Requirements after test per Design Characteristics	2488

*INDICATES MANDATORY REQUIREMENTS

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SPECIFICATION REQUIREMENT SHEET (SRS)

SHEET___ OF ___

TITLE CODE | REQUESTER NO.

1 2 3 4 5 6 7 8 9 10 11 12

THIS FORM IS USED TO INITIATE A REQUEST FOR AN AEDPS DOCUMENT (SEE PARAGRAPH TITLED SPECIFICATIONS REQUIREMENT SHEET, SECTION 5 OF MIL-STD-35).

MARK IF APERTURE CARDS REQUIRED ☐

MARK IF SEARCH ONLY ☐

REVISION TO AEDPS DOCUMENT NO.:

REQUESTER'S NAME _____

MAILING ADDRESS _____

_____

PHONE NO. A/C _____

ACTUATORS, ELECTROMECHANICAL, LINEAR
CHARACTERISTIC NAME

CHAR CODE 13 14 15 16

CHARACTERISTIC VALUE

20 25 30 35 40 45 50 55 60 65 70 75 79

LINE NO. 80

DESIGN CHARACTERISTICS - SELECT ONE OR MORE APPROPRIATE CODE(S)

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